

Python - Loop Dictionaries

Loop Through Dictionaries

Looping through dictionaries in Python refers to iterating over key-value pairs within the dictionary and performing operations on each pair. This allows you to access both keys and their corresponding values. There are several ways/methods for looping through dictionaries –

- Using a for Loop
- Using `dict.items()` method
- Using `dict.keys()` method
- Using `dict.values()` method

Loop Through Dictionary Using a For Loop

A for loop in Python is a control flow statement that iterates over a sequence of elements. It repeatedly executes a block of code for each item in the sequence. The sequence can be a range of numbers, a list, a tuple, a string, or any iterable object.

We can loop through dictionaries using a for loop in Python by iterating over the keys or key-value pairs within the dictionary. There are two common approaches –

Example: Iterating over Keys

In this approach, the loop iterates over the keys of the dictionary. Inside the loop, you can access the value corresponding to each key using dictionary indexing –



```
student = {"name": "Alice", "age": 21, "major": "Computer Science"}
for key in student:
    print(key, student[key])
```

It will produce the following output –

```
name Alice
age 21
major Computer Science
```

Example: Iterating over Key-Value Pairs

In this approach, the loop iterates over the key-value pairs using the `items()` method of the dictionary. Each iteration provides both the key and its corresponding value –



```
student = {"name": "Alice", "age": 21, "major": "Computer Science"}
for key, value in student.items():
    print(key, value)
```

We get the output as shown below –

```
name Alice
age 21
major Computer Science
```

Loop Through Dictionary Using `dict.items()` Method

The `dict.items()` method in Python is used to return a view object that displays a list of key-value pairs in the dictionary. This view object provides a dynamic view of the dictionary's items, allowing you to access both the keys and their corresponding values.

We can loop through dictionaries using the `dict.items()` method by iterating over the key-value pairs returned by this method.

Example

In this example, the `items()` method is called on the "student" dictionary, returning a view object containing the key-value pairs. The for loop iterates over each pair, assigning the key to the variable "key" and the corresponding value to the variable "value" –

```
student = {"name": "Alice", "age": 21, "major": "Computer Science"}

# Looping through key-value pairs
for key, value in student.items():
    print(key, value)
```

The output produced is as shown below –

```
name Alice
age 21
major Computer Science
```

Loop Through Dictionary Using `dict.keys()` Method

The `dict.keys()` method in Python is used to return a view object that displays a list of keys in the dictionary. This view object provides a dynamic view of the dictionary's keys, allowing you to access and iterate over them.

We can loop through dictionaries using the `dict.keys()` method by iterating over the keys returned by this method. This allows us to access and iterate over the keys of the dictionary.

Example

In the example below, the `keys()` method is called on the "student" dictionary, returning a view object containing the keys. The for loop iterates over each key in the view object, allowing you to perform operations based on the keys of the dictionary during each iteration –

```
student = {"name": "Alice", "age": 21, "major": "Computer Science"}

# Looping through keys
```

```
for key in student.keys():  
    print(key)
```

Following is the output of the above code –

```
name  
age  
major
```

Loop Through Dictionary Using dict.values() Method

The dict.values() method in Python is used to return a view object that displays a list of values in the dictionary. This view object provides a dynamic view of the dictionary's values, allowing you to access and iterate over them.

We can loop through dictionaries using the dict.values() method by iterating over the values returned by this method. This allows us to access and iterate over the values of the dictionary.

Example

In the following example, the values() method is called on the "student" dictionary, returning a view object containing the values –



```
student = {"name": "Alice", "age": 21, "major": "Computer Science"}  
  
# Looping through values  
for value in student.values():  
    print(value)
```

Output of the above code is as shown below –

```
Alice  
21  
Computer Science
```

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